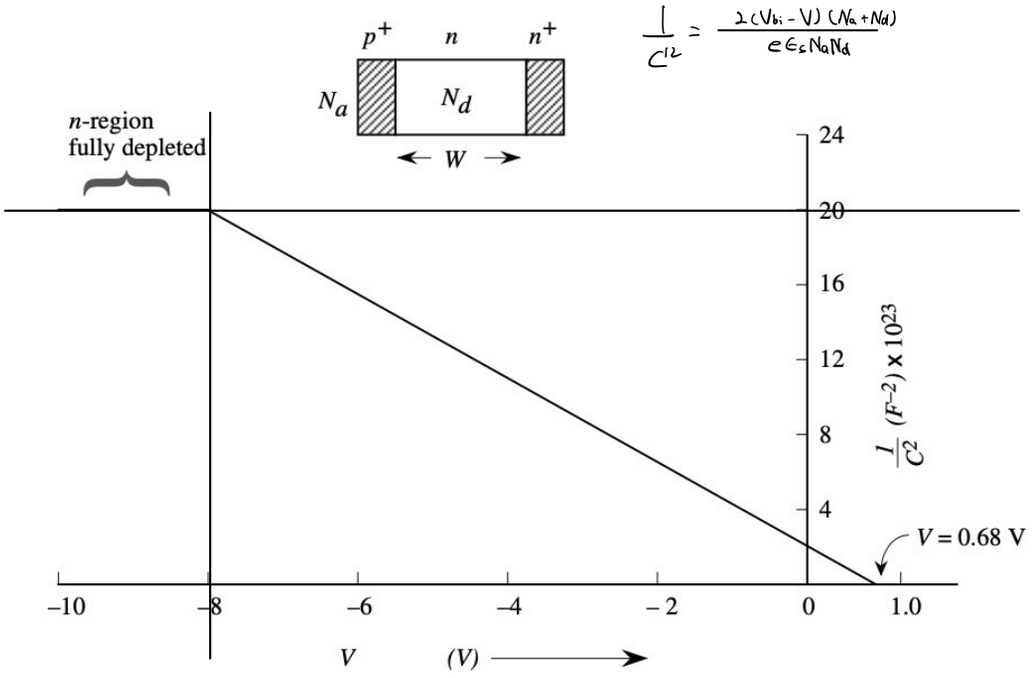
Homework 9 (due on 6/1/2026)

1. Plot the band alignment of metal-n semiconductor Schottky barrier. (10pts)
2. Plot the band alignment of metal-p semiconductor Schottky barrier. (10pts)
3. The $1/C^2$ versus applied voltage relation in a silicon p+ – n – n+ junction diode is measured to have a form shown in figure below. Calculate the thickness of the n-region, the built-in voltage, and the Na and Nd concentrations in the p+ and n regions. The diode area is 10^{-3} cm^2. Also calculate the width of the depletion. (15pts)

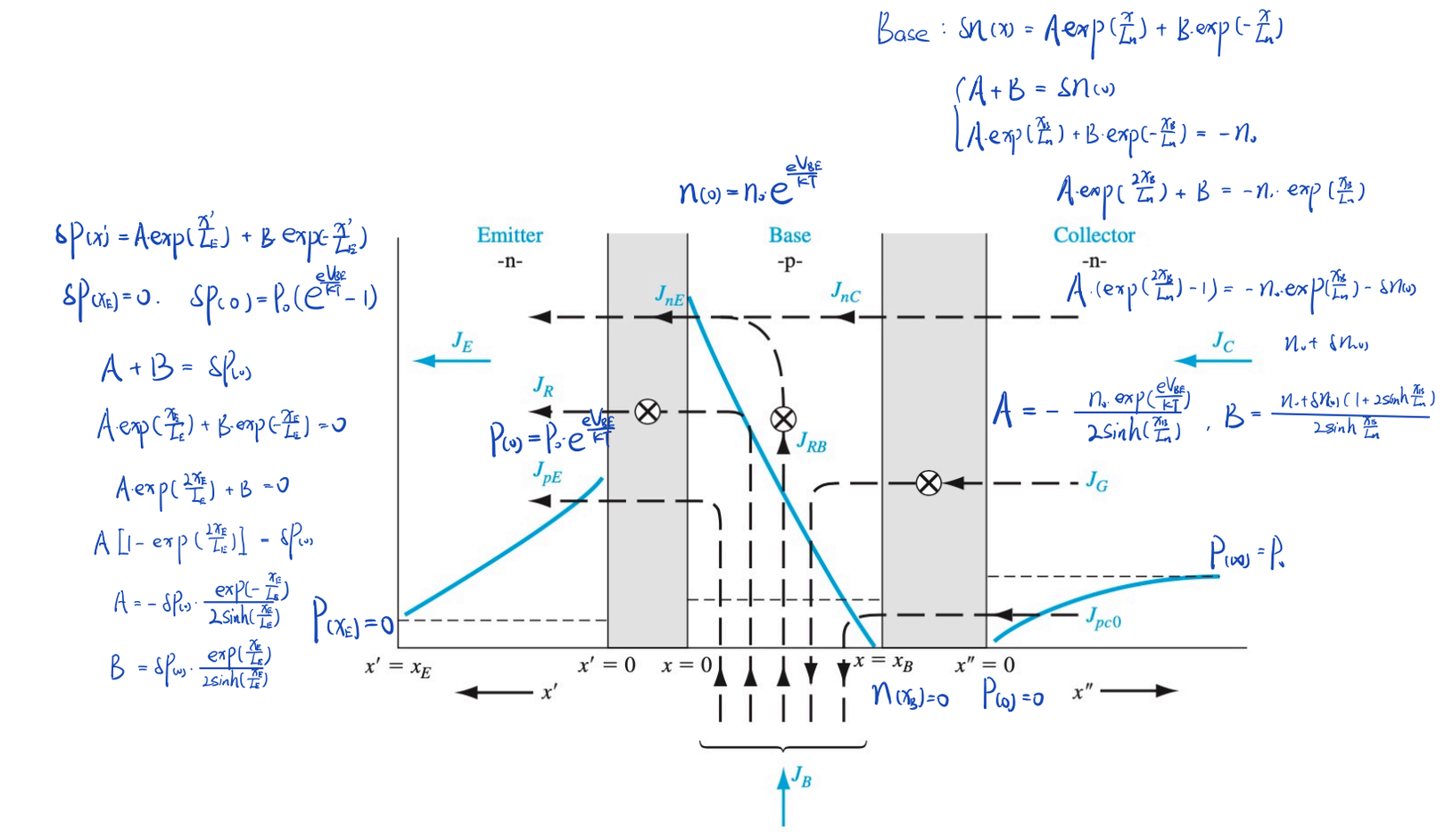


4. The capacitance of a Pt-n-type GaAs Schottky diode is given by

$$\frac{1}{(C(\mu F))^2} = 1.0 \times 10^5 - 2.0 \times 10^5 \text{ V}$$

The diode area is 0.1 cm^2. Calculate the built-in voltage Vbi, the barrier height, and the doping concentration. (15pts)

5. In the forward active mode of npn BJT,
   a) Please derive the excess minority carrier concentration in the emitter, base and collector region step by step. (15pts)
   b) Please derive the J_{nE}, J_{nC}, J_{pE} and J_{pc0} in the following picture. (10pts)
   c) Please derive and understand the formula 12.39b in Neamen's book. (10pts)



6. In the cutoff mode of npn BJT, please derive the excess minority carrier concentration in the emitter, base and collector region step by step. (15pts)

Cut off.

Emitter : $\delta p(x') = A \cdot \exp\left(\frac{x'}{L_n}\right) + B \cdot \exp\left(-\frac{x'}{L_n}\right)$

$\delta p(\infty) = -p_0 \quad . \quad \delta p(x_E) = 0 .$

$\Rightarrow A + B = -p_0 . \qquad A \cdot \exp\left(\frac{x_E}{L_n}\right) + B \cdot \exp\left(-\frac{x_E}{L_n}\right) = 0$

$A \cdot \left[\exp\left(\frac{x_E}{L_n}\right) - 1\right] = p_0 .$

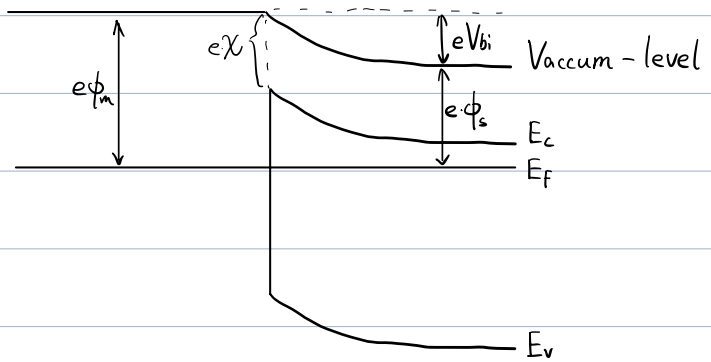
$$A = \frac{p_0}{\exp\left(\frac{2x_E}{L_n}\right) - 1} = \frac{p_0 \cdot \exp\left(-\frac{x_E}{L_n}\right)}{2\sinh\left(\frac{x_E}{L_n}\right)}$$

$$B = -p_0 - A = -p_0 \cdot \frac{\exp\left(\frac{2x_E}{L_n}\right)}{\exp\left(\frac{2x_E}{L_n}\right) - 1} = -p_0 \cdot \frac{\exp\left(\frac{x_E}{L_n}\right)}{2\sinh\left(\frac{x_E}{L_n}\right)}$$

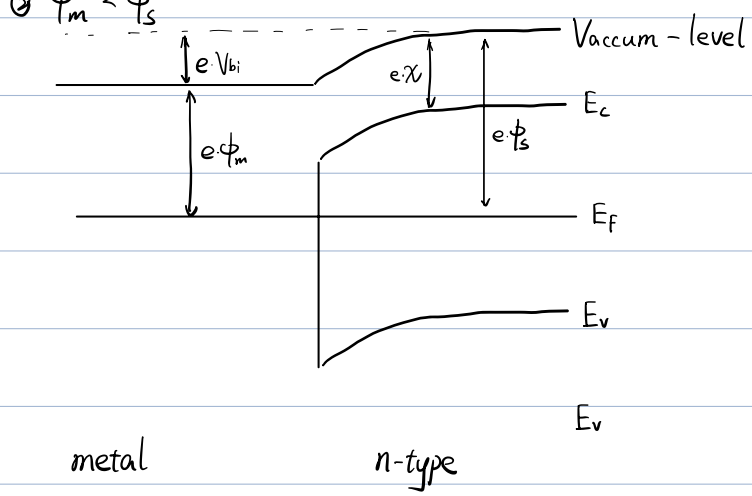
$$\delta p(x') = \frac{p_0}{2\sinh\left(\frac{x_E}{L_n}\right)} \cdot \exp\left(\frac{x' - x_E}{L_n}\right) - \frac{p_0}{2\sinh\left(\frac{x_E}{L_n}\right)} \cdot \exp\left(\frac{-x' + x_E}{L_n}\right)$$

1.

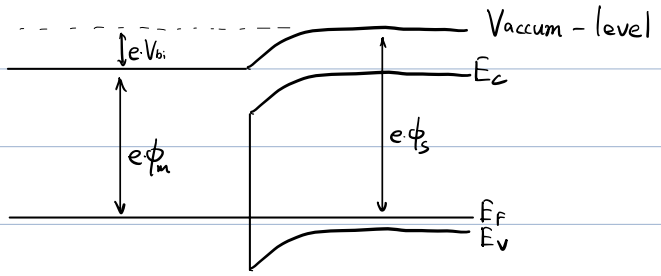
① $\phi_m > \phi_s$



② $\phi_m < \phi_s$

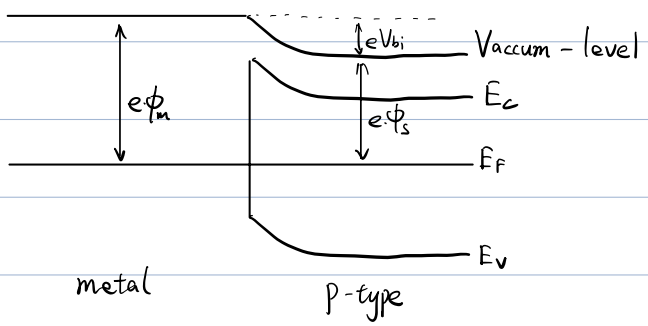


2. ① $\phi_m < \phi_s$



metal p-type

② $\phi_m > \phi_s$



metal p-type

3. $W = x_p + x_n = \left\{ \frac{2\epsilon_s (V_{bi} - V)}{e} \cdot \left[\frac{N_a + N_d}{N_a N_d} \right] \right\}^{\frac{1}{2}}$

$C = \frac{\epsilon_s A}{W}$

$\Rightarrow \left(\frac{1}{C}\right)^2 = \frac{2(V_{bi} - V)}{e \epsilon_s A^2} \cdot \frac{N_a + N_d}{N_a N_d}$

slop: $k = -2.304 \times 10^{23}\ F^{-2}/V$, $V_{bi} = 0.68\ V$

take $\epsilon_s = 1.04 \times 10^{-12}\ F/cm$, $A = 10^{-3}\ cm^2$ $\Rightarrow \frac{N_a + N_d}{N_a N_d} \overset{N_a \gg N_d}{=\!=} \frac{1}{N_d} = 1.92 \times 10^{-19}\ cm^3$

$V_{bi} = \frac{kT}{e} \ln \frac{N_a N_d}{n_i^2}$, take $\frac{kT}{e} = 0.026\ V$, $n_i = 10^{10}\ cm^{-3}$ $\Rightarrow N_a N_d = 2.28 \times 10^{31}\ cm^{-6}$

while $V \leq -8V$, n-region fully depleted.

$\Rightarrow$ thickness of n-region: $W_n = 1.471 \times 10^{-3}\ cm = 14.71\ \mu m$

$\Rightarrow V_{bi} = 0.68\ V$

$\Rightarrow N_a = 4.38 \times 10^{17}\ cm^{-3}$. $N_d = 5.20 \times 10^{13}\ cm^{-3}$

4. $\frac{1}{(C(\mu F))^2} = 1.0 \times 10^5 - 2.0 \times 10^5\ V$

$\left(\frac{1}{C}\right)^2 = \frac{2(V_{bi} - V)}{e \epsilon_s A^2} \cdot \frac{1}{N_D}$, take $\epsilon_s = 1.16 \times 10^{-12}\ F/cm$

$\Rightarrow V_{bi} = 0.5\ V$

$\frac{2}{e \epsilon_s A^2} \cdot \frac{1}{N_D} = 2 \times 10^5\ (\mu F)^{-2}/V = 2 \times 10^{17}\ F^{-2}/V$

$\Rightarrow N_D = 5.38 \times 10^{15}\ cm^{-3}$

$\phi_n = \frac{kT}{e} \ln \frac{N_c}{N_D}$, take $T = 300k$, $N_c = 4.7 \times 10^{17}\ cm^{-3}$

Barrier height: $\phi_b = V_{bi} + \phi_n$

$\Rightarrow \phi_b = 0.62\ V$

5. (a) Emitter : $D_E \cdot \frac{\partial^2(\delta P_E(x'))}{\partial x'^2} - \frac{\delta P_E(x')}{\tau_{E0}} = 0$

$\Rightarrow \delta P_E(x') = A\exp(\frac{x'}{L_E}) + B\exp(\frac{-x'}{L_E})$ $L_E = \sqrt{D_E \cdot \tau_{E0}}$

Boundary condition: $\delta P_E(x_E) = A\exp(\frac{x_E}{L_E}) + B\exp(-\frac{x_E}{L_E}) = 0$

$\delta P_E(0) = A + B = P_{E0}[\exp(\frac{eV_{BE}}{kT}) - 1]$

$\Rightarrow A = -\frac{\delta P_E(0)\exp(-\frac{x_E}{L_E})}{2\sinh(\frac{x_E}{L_E})}$, $B = \frac{\delta P_E(0)\exp(\frac{x_E}{L_E})}{2\sinh(\frac{x_E}{L_E})}$

$\Rightarrow \delta P_E(x') = \delta P_E(0)\cdot\frac{\sinh(\frac{x_E - x'}{L_E})}{\sinh(\frac{x_E}{L_E})} = \frac{P_{E0}[\exp(\frac{eV_{BE}}{kT}) - 1]\cdot\sinh(\frac{x_E - x'}{L_E})}{\sinh(\frac{x_E}{L_E})}$

Base : $D_B \cdot \frac{\partial^2(\delta n_B(x))}{\partial x^2} - \frac{\delta n_B(x)}{\tau_{B0}} = 0$, $\delta n_B(x) = A\cdot\exp(\frac{x}{L_B}) + B\cdot\exp(\frac{-x}{L_B})$, $L_B = \sqrt{D_B\tau_{B0}}$

Boundary condition: $\delta n_B(0) = n_B(0) - n_{B0} = n_{B0}[\exp(\frac{eV_{BE}}{kT}) - 1]$

$\delta n_B(x_B) = n_B(0) - n_{B0} = -n_{B0}$

$\Rightarrow A = \frac{\delta n_B(x_B) - \delta n_B(0)\exp(-\frac{x_B}{L_B})}{2\sinh(\frac{x_B}{L_B})}$, $B = \frac{\delta n_B(0)\exp(\frac{x_B}{L_B}) - \delta n_B(x_B)}{2\sinh(\frac{x_B}{L_B})}$

$\Rightarrow \delta n_B(x) = \frac{n_{B0}\{[\exp(\frac{eV_{BE}}{kT}) - 1]\sinh(\frac{x_B - x}{L_B}) - \sinh(\frac{x}{L_B})\}}{\sinh(\frac{x_B}{L_B})}$

Consider $x_B \ll L_B$ $\delta n_B(x) = \frac{n_{B0}}{x_B}\{[\exp(\frac{eV_{BE}}{kT}) - 1](x_B - x) - x\}$

Collector : $D_C \cdot \frac{\partial^2(\delta P_C(x''))}{\partial x''^2} - \frac{\partial P_C(x')}{\tau_{C0}} = 0$, $\delta P_C(x'') = A\cdot\exp(\frac{x''}{L_C}) + B\cdot\exp(\frac{-x''}{L_C})$, $L_C = \sqrt{D_C\tau_{C0}}$

Boundary condition: $\delta P_C(x_C) = 0$ $\delta P_C(0) = P_C(0) - P_{C0} = -P_{C0}$

$\Rightarrow A\cdot\exp(\frac{x_C}{L_C}) + B\exp(-\frac{x_C}{L_C}) = 0$, $A + B = -P_{C0}$

$\Rightarrow \delta P_C(x'') = -P_{C0}\cdot\exp(-\frac{x''}{L_C})$

(b) $\delta n_B(x) = \dfrac{n_{B0}\left\{\left[\exp\left(\frac{eV_{BE}}{kT}\right)-1\right]\sinh\left(\frac{x_B-x}{L_B}\right)-\sinh\left(\frac{x}{L_B}\right)\right\}}{\sinh\left(\frac{x_B}{L_B}\right)}$ $\qquad \delta p_E(x') = \dfrac{p_{E0}\left[\exp\left(\frac{eV_{BE}}{kT}\right)-1\right]\sinh\left(\frac{x_E-x'}{L_E}\right)}{\sinh\left(\frac{x_E}{L_E}\right)}$

① $J_{nE} = -eD_E\left.\dfrac{d[\delta n_B(x)]}{dx}\right|_{x=0} = \dfrac{eD_E\cdot n_{B0}\left\{\left[\exp\left(\frac{eV_{BE}}{kT}\right)-1\right]\cosh\left(\frac{x_B}{L_B}\right)+1\right\}}{L_B\cdot\sinh\left(\frac{x_B}{L_B}\right)}$

② $J_{nC} = -eD_B\left.\dfrac{d[\delta n_B(x)]}{dx}\right|_{x=x_B} = \dfrac{eD_E\cdot n_{B0}\left\{\left[\exp\left(\frac{eV_{BE}}{kT}\right)-1\right]+\cosh\left(\frac{x_B}{L_B}\right)\right\}}{L_B\cdot\sinh\left(\frac{x_B}{L_B}\right)}$

③ $J_{pE} = -eD_E\left.\dfrac{d[\delta p_E(x')]}{dx'}\right|_{x'=0} = \dfrac{eD_E\,p_{E0}\left[\exp\left(\frac{eV_{BE}}{kT}\right)-1\right]\cosh\left(\frac{x_E}{L_E}\right)}{L_E\cdot\sinh\left(\frac{x_E}{L_E}\right)}$

④ $J_{pC_0} = eD_C\left.\dfrac{d[\delta p_C(x'')]}{dx''}\right|_{x''=0} = \dfrac{eD_C\,p_{C0}}{L_C}$

(c) $\alpha_T = \dfrac{J_{nC}}{J_{nE}} = \dfrac{\exp\left(\frac{eV_{BE}}{kT}\right)-1+\cosh\left(\frac{x_B}{L_B}\right)}{\left[\exp\left(\frac{eV_{BE}}{kT}\right)-1\right]\cdot\cosh\left(\frac{x_B}{L_B}\right)+1}$

typically, $\exp\left(\frac{eV_{BE}}{kT}\right) \gg 1$.

$\Rightarrow \alpha_T \approx \dfrac{\exp\left(\frac{eV_{BE}}{kT}\right)+\cosh\left(\frac{x_B}{L_B}\right)}{\exp\left(\frac{eV_{BE}}{kT}\right)\cdot\cosh\left(\frac{x_B}{L_B}\right)+1}$

for $x_B \ll L_B$, $\exp\left(\frac{eV_{BE}}{kT}\right) \gg \cosh\left(\frac{x_B}{L_B}\right)$

$\Rightarrow \alpha_T \approx \dfrac{1}{\cosh\left(\frac{x_B}{L_B}\right)+1} \approx \dfrac{1}{1+\frac{1}{2}\left(\frac{x_B}{L_B}\right)^2} \approx 1-\frac{1}{2}\left(\frac{x_B}{L_B}\right)^2$

6. Emitter :

$$D_E \cdot \frac{d^2 \delta P_E(x')}{dx'^2} - \frac{\delta P_E(x')}{\tau_{E0}} = 0$$

$$\Rightarrow \delta P_E(x') = A e^{\frac{x'}{L_E}} + B e^{-\frac{x'}{L_E}}, \quad L_E = \sqrt{D_E \tau_{E0}}$$

Boundary condition: $\delta P_E(x_E) = 0$, $\delta P_E(0) = -P_{E0}$

$$A = \frac{-P_{E0}}{1 - \exp(\frac{2x_E}{L_E})}, \quad B = \frac{-P_{E0} \exp(\frac{x_E}{L_E})}{2\sinh(\frac{x_E}{L_E})}$$

$$\Rightarrow \delta P_E(x') = \frac{-P_{E0} \cdot \sinh(\frac{x_E - x'}{L_E})}{\sinh(\frac{x_E}{L_E})}$$

Base : $D_B \cdot \frac{\partial^2(\delta n_B(x))}{\partial x^2} - \frac{\partial n_B(x)}{\tau_{B0}} = 0$, $\delta n_B(x) = A \cdot \exp(\frac{-x}{L_B}) + B \cdot \exp(\frac{-x}{L_B})$, $L_B = \sqrt{D_B \tau_{B0}}$

Boundary condition: $\delta n_B(0) = -n_{B0}$

$$\delta n_B(x_B) = -n_{B0}$$

$$\Rightarrow A = B = \frac{-n_{B0}}{2\cosh(\frac{x_B}{2L_B})} \cdot \exp(\frac{-x_B}{2L_B})$$

$$\Rightarrow \delta n_B(x) = -n_{B0} \frac{\cosh(\frac{x_B - 2x}{2L_B})}{\cosh(\frac{x_B}{2L_B})}$$

Collector : $D_c \cdot \frac{\partial^2(\delta P_c(x''))}{\partial x''^2} - \frac{\partial P_c(x'')}{\tau_{c0}} = 0$, $\delta P_c(x'') = A \cdot \exp(\frac{-x''}{L_c}) + B \cdot \exp(\frac{-x''}{L_c})$, $L_c = \sqrt{D_c \tau_{c0}}$

Boundary condition: $\delta P_c(x_c) = 0$, $\delta P_c(0) = P_c(0) - P_{c0} = -P_{c0}$

$$\Rightarrow A \cdot \exp(\frac{x_c}{L_c}) + B \exp(-\frac{x_c}{L_c}) = 0, \quad A + B = -P_{c0}$$

$$\Rightarrow \delta P_c(x'') = -P_{c0} \exp(-\frac{x''}{L_c})$$